

MON And The Kids

Attendance and Leave Management System

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Khon Kaen University
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Chapter 1 - Introduction

1.1 Product Information

Product Name	MON And The Kids ; Attendance and Leave Management System
Date	March 17, 2026
Functional & Non-Functional requirement	Functional & Non-functional requirement
Lark links	Backlog : Agile
Docs links	Member & Contribution
Figma (Sprint 1&2)	Agile - Figma
Customer	A+

1.2 Background of the Project

MON And The Kids – Attendance and Leave Management System is a web-based system designed to replace manual attendance and leave processes with a centralized digital solution. It supports three user roles ; Employee, Manager, HR with role-specific access and features including check-in/check-out tracking, leave request and approval, employee management, reporting, and notifications. The system is developed using Agile methodology across two sprints: Sprint 1 covers authentication and role-based access control, and Sprint 2 covers attendance and leave management.

1.3 Objectives of the Project

This project aims to develop a web-based attendance and leave management system with role-based access control, automated check-in/check-out with late detection, an end-to-end leave request workflow, centralized HR management with

reporting, and non-functional standards including under 2-second response time, 99.5% uptime, bilingual support, and secure data handling.

1.4 Scope of the System

MON And The Kids, Attendance and Leave Management System covers the following functional scope :

- User Management : Login/logout with role-based access for Officer, Manager, HR, and Admin roles.
- Employee Information Management : HR can add, edit, and view all employee records; Managers and Officers have view access according to their role.
- Attendance System : Daily check-in/check-out with timestamp recording, late arrival detection, and attendance history viewing.
- Leave Management System : Leave request submission, approval/rejection workflow, leave balance tracking, status notifications, and rejection reason display.
- Resource Management : Calendar view of team and personal leave dates, daily check-in overview for planning purposes.
- Reporting System : Personal, department, and organization-level reports with Excel export capability.
- Notification System : In-app and/or email alerts for late check-ins, missed check-outs, and leave request updates.

Chapter 2 - Requirements and Product Definition

2.1 Personas

Photo			
Name	Ploy Supaporn Wongchai	Chai Somchai Rattanakul	Nat Nattapon Srisawat
Age	32	42	27
Role	HR Manager	Department Manager	Employee (Officer)
Occupation	HR Manager	Marketing Department Manager	Marketing Officer
Personality	Detail-oriented, systematic, multitasker	Busy, decisive, meeting-driven	Tech-savvy, efficiency-focused, punctual
Tech Savviness	Moderate – High	Moderate	High
Problems	Paper-based forms get lost, manual leave calculation, no real-time attendance visibility, data scattered in Excel	Unaware of team absences until last minute, misses approval requests during meetings, hard to plan projects	Unsure if request was received, long wait for approval, forgets to check-in/out
Expectations	Dashboard with pending requests, auto leave balance, one-click reports, real-time team calendar	Clear notifications, team calendar view, simple approval process, daily staffing count	Prominent check-in/out button, leave balance on dashboard, status tracking, check-out reminder
Key Features Used	Leave approval, employee management, reports, leave policy	Leave approval, team calendar, attendance reports, notifications	Check-in/out, leave request, leave balance, attendance history

	settings, notifications		
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2.2 WOW Factor of the Project

Sprint 1

Sprint 1 focused on building the foundation of the MON And The Kids, Attendance and Leave Management System, secure login and role-based access control. While authentication may seem like a standard feature, our implementation goes beyond a simple username/password check by delivering a fully personalized and context-aware experience for every user.

Smart Role-Based Homepage Redirect

- **What it does :** After logging in, the system automatically identifies the user's role and redirects them to a customized homepage designed specifically for that role — HR, Manager, or Officer without any extra steps
- **How it works :** Upon successful login, the system retrieves the user's role from the database and dynamically loads the corresponding interface. Each homepage displays only the features, menus, and data relevant to that role. An HR user sees employee management tools and pending approvals; a Manager sees their team's attendance overview; an Officer sees their own check-in status and leave balance
- **Why it stands out :** Most basic systems show the same interface to all users and rely on permission flags to hide irrelevant menus. Our system takes a different approach, each role gets a fully tailored dashboard that reduces visual clutter, improves usability, and ensures users can act immediately upon login without navigating through unrelated features.

Intelligent Failed Login Handling

- **What it does :** The system handles login failures gracefully by providing clear, specific error messages
- **How it works :** When a user enters incorrect credentials, the system detects whether the username, password, or both are invalid and displays an

appropriate error message. If the username does not exist, an 'Invalid username' message is shown. If the password is wrong, an 'Invalid password' message appears. If both fields are left empty, the system prompts the user to fill in all required fields. The user remains on the login page and can retry immediately.

- **Why it stands out :** Rather than showing a generic 'Login failed' message for all error types, which is common in basic systems. Our implementation gives users actionable feedback so they can correct their input quickly. This reduces user frustration and support requests while maintaining system security.

Sprint 2

Sprint 2 introduced the core daily-use features of the system — attendance check-in/check-out and leave request management. The WOW factors in this sprint focus on smart UI controls and real-time feedback that make the system intuitive and error-resistant for everyday use.

Context-Aware Check-In / Check-Out Button States

- **What it does :** The Check-In and Check-Out buttons are not static — they dynamically enable or disable themselves based on the employee's current context, preventing invalid actions before they happen.
- **How it works :** The system evaluates three conditions in real time:
 - whether the current time falls within the allowed check-in or check-out window,
 - whether the employee has already checked in or checked out for the day, and
 - whether the employee's location is within the allowed company area.

If any condition is not met, the button is automatically disabled and cannot be clicked. For example, if an employee has already checked in, the Check-In button becomes disabled immediately — and the Check-Out button remains disabled until the allowed check-out time.

- **Why it stands out :** A typical attendance system relies on server-side validation after the user submits an action, meaning users can attempt

invalid check-ins and only receive an error after the fact. Our approach disables the button proactively — the UI itself communicates what is and isn't allowed at any given moment. This eliminates accidental duplicate entries, reduces error messages, and creates a much smoother daily experience for employees.

Real-Time Leave Balance Display Before Submission

- **What it does :** When an employee opens the leave request form, their remaining leave balance is displayed immediately — before they fill in any details — so they can make informed decisions without navigating away.
- **How it works :** The system retrieves the employee's leave entitlement and deducts any approved or pending leave days to calculate the current available balance. This balance is shown prominently on the leave request page broken down by leave type (e.g., Sick Leave, Annual Leave, Personal Leave). As the employee selects leave type and enters the duration, the system updates the display dynamically to show how many days would remain after the request.
- **Why it stands out :** In many systems, employees submit leave requests and only discover they have insufficient balance after the request is rejected. This creates unnecessary back-and-forth between employees, managers, and HR. By surfacing the leave balance at the point of decision — and updating it dynamically as the form is filled — our system empowers employees to self-manage their leave without surprises, and reduces the volume of invalid submissions that HR and managers have to handle.

2.3 Product Backlog

The product backlog was revised to align with the project activities represented in the PERT diagram and Critical Path Analysis. In addition to system features, technical tasks such as UI design, system modeling (DFD and ER diagrams), testing, and integration were included because they are essential for delivering functional increments in each sprint.

2.3.1 Product Backlog Item

ID	Backlog Item	Type	Priority	Estimate (SP)	Planned Sprint
PB-01	Design UI Login Page	Task	High	1	Sprint 1
PB-02	Design UI Failed Login Notification	Task	High	1	Sprint 1
PB-03	Design UI Role-Based Homepage (HR / Manager / Officer)	Task	High	1	Sprint 1
PB-04	Design DFD for Authentication System	Task	High	1	Sprint 1
PB-05	Design ER Diagram for Authentication Data	Task	High	1	Sprint 1
PB-06	Create Test Cases for Login / Failed Login	Task	High	1	Sprint 1
PB-07	Create Test Cases for Role-Based Homepage	Task	Medium	2	Sprint 1
PB-08	Improve UI Based on Sprint 1 Feedback	Task	High	1	Sprint 2
PB-9	Design UI Check-in / Check-out Page	Task	High	1	Sprint 2

PB-10	Design UI Leave Request Page	Task	High	1	Sprint 2
PB-11	Design DFD for Attendance and Leave Request System	Task	High	1	Sprint 2
PB-12	Design ER Diagram for Attendance and Leave Data	Task	Medium	1	Sprint 2
PB-13	Create Test Cases for Check-in / Check-out	Task	High	2	Sprint 2
PB-14	Create Test Cases for Leave Request	Task	High	2	Sprint 2
PB-15	Improve UI / Fix Issues for Sprint 2 Approval Workflow	Task	Medium	1	Sprint 2

2.3.2 User Stories

No.	Persona	User Requirement	User Story	Acceptance Criteria
1	System User	US001 : User Authentication & RBAC	As a system user, I want to log in securely and access features based on my role, so that data is protected and I can use only authorized functions.	- Login with username and password - Role-based access: Officer accesses personal functions only - Role-based access: Manager accesses team data + personal functions - Role-based access: HR accesses all employee data + management functions - Role-based access: Admin accesses full system configuration - Session timeout and secure logout - Failed login lockout after repeated attempts
2	HR Officer	US002 : Employee Information Management	As an HR officer, I want to manage employee information in one centralized system, so that employee records are accurate and up to date.	- HR can Add / Edit / View all employee records - Manager can view personal info and subordinate info within their department - Officer can view their own personal info only (name, dept, position, leave entitlements, attendance)

3	Employee	US003 : Attendance Management System	As an employee, I want to check in and check out daily so that my attendance is recorded	- All roles can perform daily check-in / check-out with timestamp - All roles can view own check-in/check-out confirmation status - All roles can view own attendance history and late arrival records - Manager can view attendance status and late arrival statistics of team members - HR can review and monitor attendance records of all employees - System flags late arrivals automatically
4	Employee	US004 : Leave Management System	As an Employee, I want to submit a leave request so that I can request time off.	- All roles can submit leave request (select leave type, select start/end date, provide reason) - All roles can view remaining leave balance before submission - All roles can edit leave request before final submission - All roles can view status of leave requests (Pending / Approved / Rejected) - All roles can view rejection reason when request is rejected - All roles receive notification of leave approval results - Manager can Approve / Reject team members' leave requests with optional comment

				- HR can Approve / Reject all employee leave requests with optional comment - Manager and HR receive notification when a leave request is submitted
5	Manager	US005 : Resource & Workforce Planning	As a manager, I want to view team availability on a calendar, so that I can plan workforce resources effectively.	- Officer can view personal leave dates and scheduled event dates on calendar - Manager can view team calendar (who is on leave / who is working) - Manager can view team daily check-in overview - HR can view all employee calendar and daily check-in overview
6	HR Officer	US006 : Reporting & Analytics	As an HR officer, I want to generate attendance and leave reports, so that management can make data-driven decisions.	- Officer can view personal attendance and leave reports - Manager can view department summary reports (late arrivals, absences, leave) - Manager can export department summary reports to Excel - HR can generate annual attendance and leave statistics reports - HR can export organization-wide summary reports to Excel
7	System User	US007 : Notification & Alert System	As a system user, I want to receive important notifications and alerts, so that I can take action promptly.	- All roles receive notification for late check-ins or missed check-outs - Manager receives notification when a team member submits a leave request - HR receives notification when any employee submits a leave request

				- Notifications delivered in-app and / or via email
8	User	US008 : System Quality, Performance & Localization	As a user, I want the system to be fast, reliable, and available in Thai and English, so that I can use it efficiently anytime.	- System response time does not exceed 2 seconds for normal operations - Check-in and check-out actions are processed immediately - System uptime is at least 99.5% - Supports Thai and English languages (toggle for all roles) - Role-based access control (RBAC) enforced throughout - All sensitive data encrypted in the database - User passwords stored in hashed format - System maintains data consistency during concurrent access - System allows future expansion of features

Chapter 3 - System Design

3.1 Sprint Planning (2 Sprints)

Sprint 1

- Sprint Duration : 1 Week
- Sprint Goal : To design login and access features based on role with their system flows

Selected Backlog Items :

ID	Backlog items	Estimate(SP)
PB-01	Design UI Login Page	1
PB-02	Design UI Failed Login Notification	1
PB-03	Design UI Role-Based Homepage (HR / Manager / Officer)	1
PB-04	Design DFD for Authentication System	1
PB-05	Design ER Diagram for Authentication Data	1
PB-06	Create Test Cases for Login / Failed Login / Forgot Password	1
PB-07	Create Test Cases for Role-Based Homepage	2
PB-08	Improve UI Based on Sprint 1 Feedback	1

Sprint 2

- Sprint Duration : 1 Week
- Sprint Goal : To design the check in, check out and leave result pages with their system flows

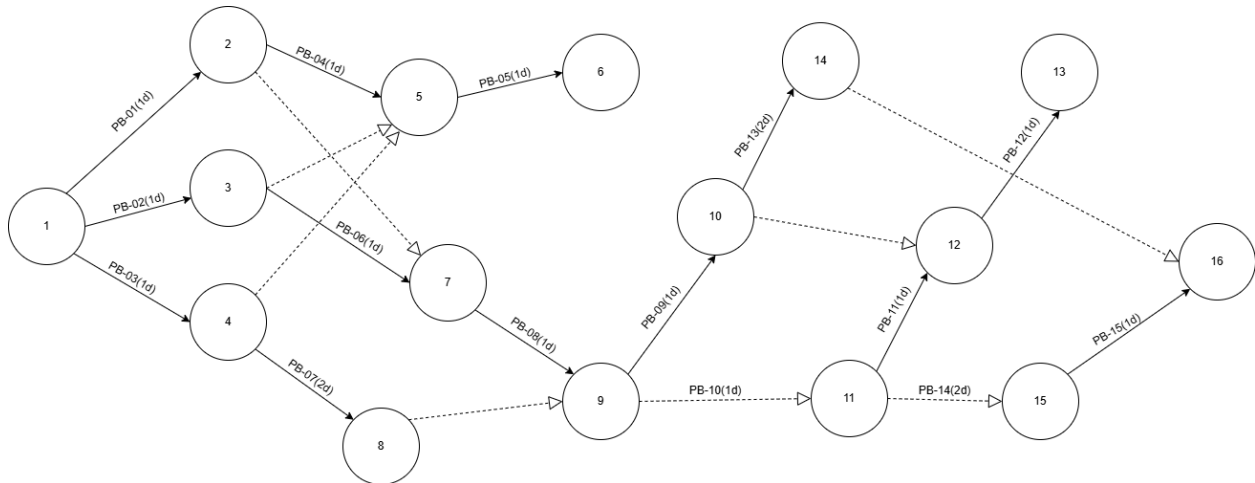
Selected Backlog Items :

ID	Backlog items	Estimate(SP)
PB-9	Design UI Check-in / Check-out Page	1
PB-10	Design UI Leave Request Page	1
PB-11	Design DFD for Attendance and Leave Request System	1
PB-12	Design ER Diagram for Attendance and Leave Data	1
PB-13	Create Test Cases for Check-in / Check-out	1
PB-14	Create Test Cases for Leave Request	2

PB-15	Improve UI / Fix Issues for Sprint 2	2
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3.2 PERT Diagram

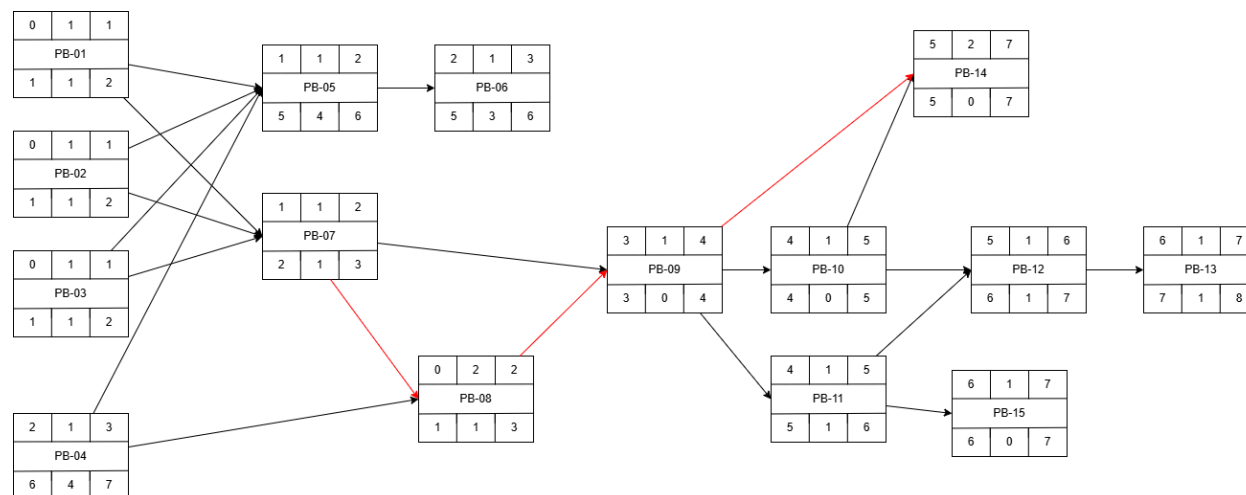
ID	Activity	Duration (days)	Predecessor
PB-01	Design UI Login Page	1	-
PB-02	Design UI Failed Login Notification	1	-
PB-03	Design UI Role-Based Homepage (HR / Manager / Officer)	1	-
PB-04	Design DFD for Authentication System	1	PB-01, PB-02, PB-03
PB-05	Design ER Diagram for Authentication Data	1	PB-04
PB-06	Create Test Cases for Login / Failed Login / Forgot Password	1	PB-01, PB-02
PB-07	Create Test Cases for Role-Based Homepage	2	PB-03
PB-08	Improve UI Based on Sprint 1 Feedback	1	PB-06, PB-07
PB-09	Design UI Check-in / Check-out Page	1	PB-08
PB-10	Design UI Leave Request Page	1	PB-08
PB-11	Design DFD for Attendance and Leave Request System	1	PB-9, PB-10
PB-12	Design ER Diagram for Attendance and Leave Data	1	PB-11
PB-13	Create Test Cases for Check-in / Check-out	2	PB-9
PB-14	Create Test Cases for Leave Request	2	PB-10
PB-15	Improve UI / Fix Issues for Sprint 2	1	PB-13, PB-14



3.3 Critical Path Analysis

Activity	Dur	ES	EF	LS	LF	TF
PB-01	1	0	1	0	1	0
PB-02	1	0	1	0	1	0
PB-03	1	1	2	5	6	4
PB-04	1	2	3	6	7	4
PB-05	1	1	2	1	2	0
PB-06	2	0	2	0	2	0
PB-07	1	2	3	2	3	0
PB-08	1	3	4	3	4	0
PB-09	1	3	4	3	4	0
PB-10	1	4	5	5	6	1
PB-11	1	5	6	6	7	1
PB-12	2	4	6	4	6	0
PB-13	2	4	6	4	6	0
PB-14	1	6	7	6	7	0

PB-15	1	6	7	6	7	0
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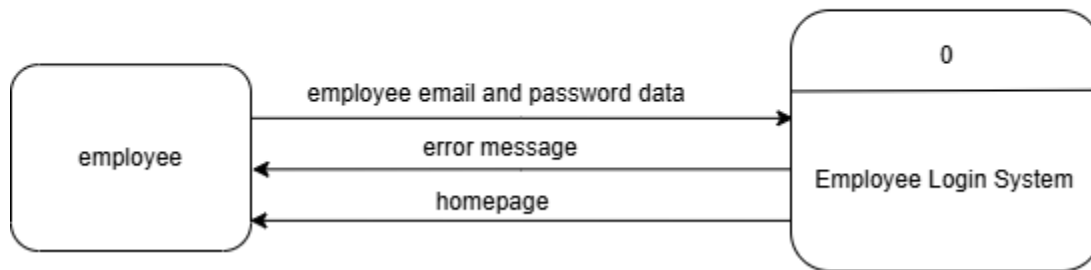
Critical path : PB-07 → PB-08 → PB-09 → PB-14

Total project duration : 7 days

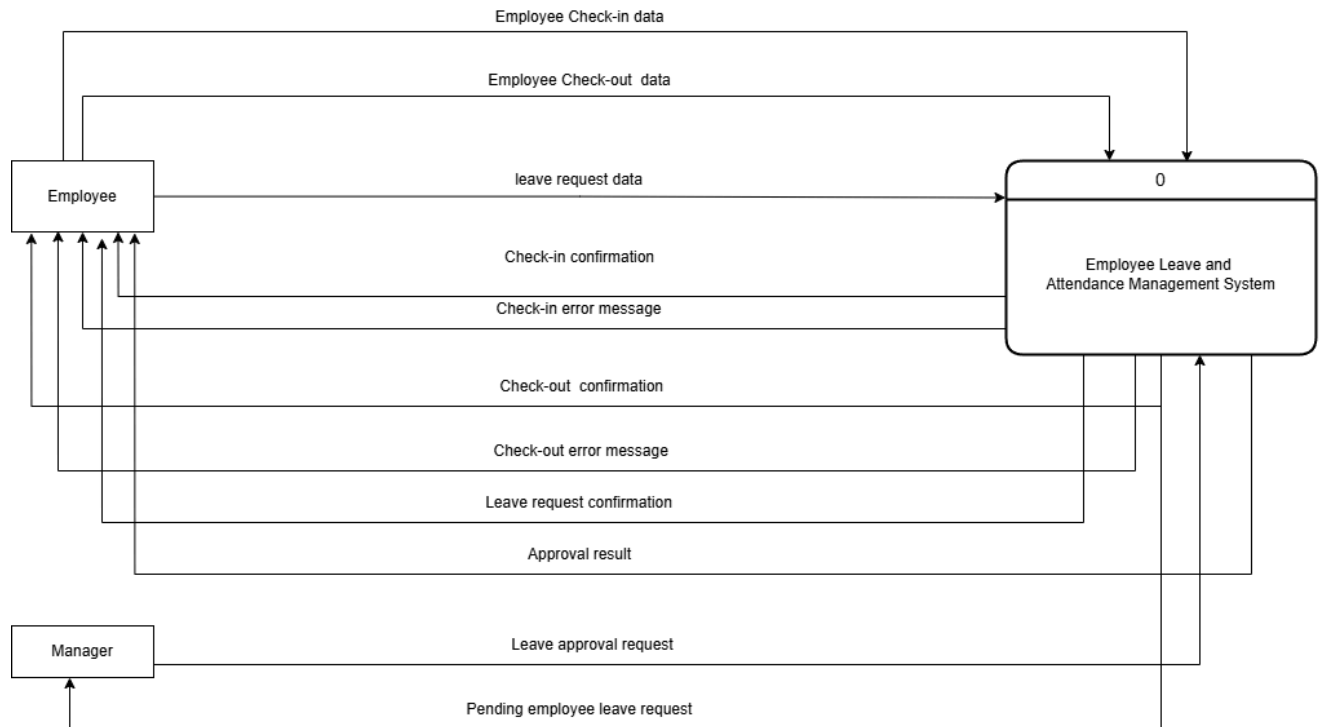
Chapter 4 - System Design and Analysis

4.1 Context Diagram

4.1.1 Sprint 1



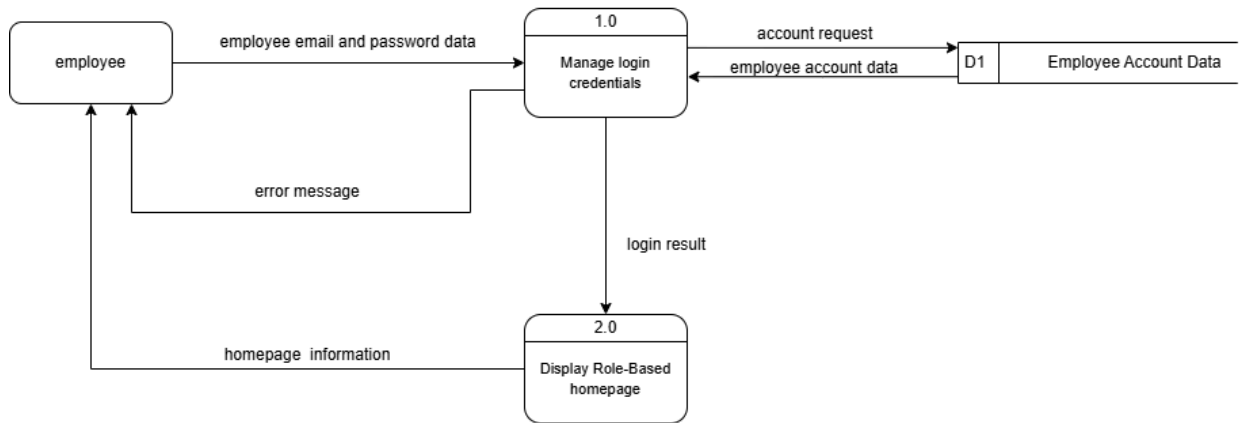
4.1.2 Sprint 2



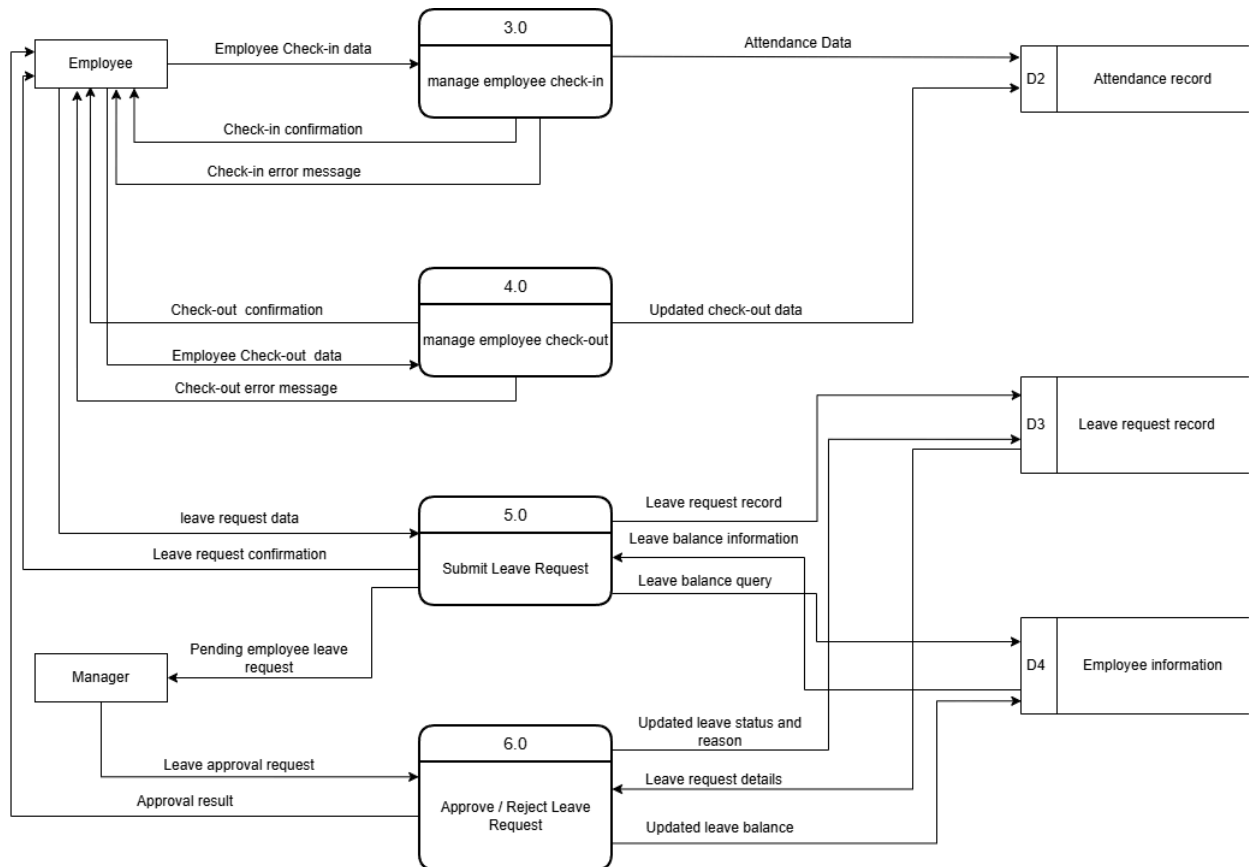
4.2 Data Flow Diagram (DFD)

4.2.1 DFD Level 1

- Sprint 1 [DFD ER Sprint 1 Draw.io](#)

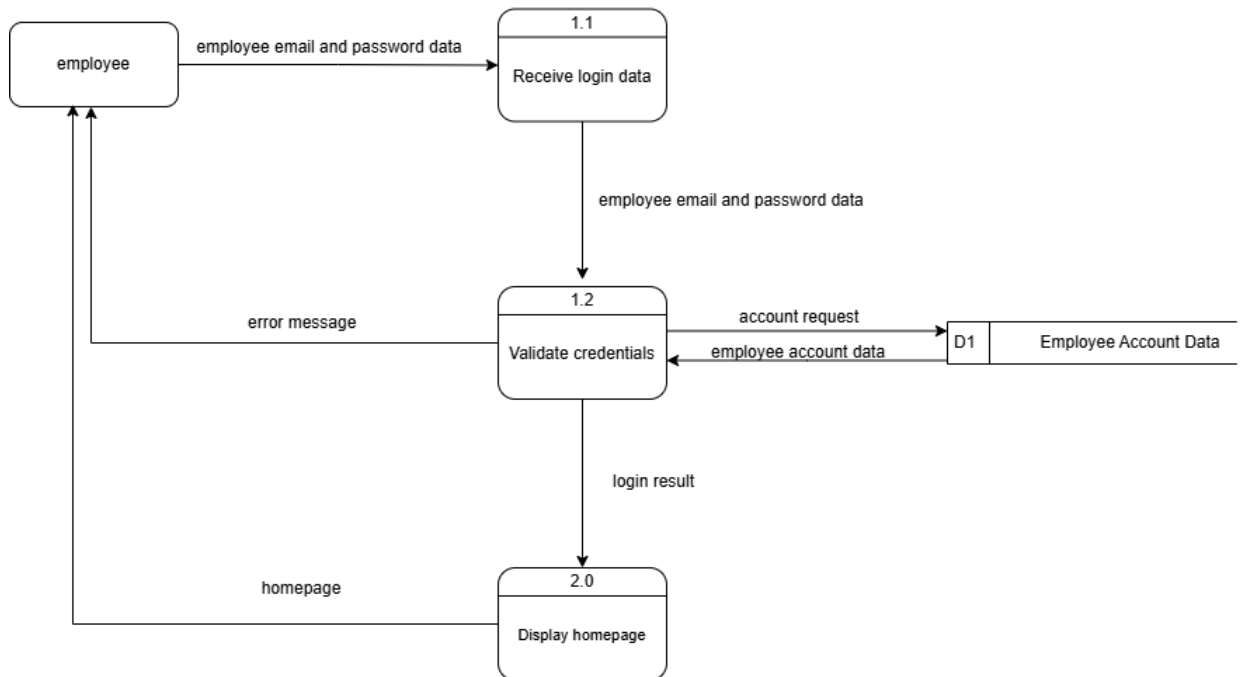


- Sprint 2 [DFD ER Sprint 2 Draw.io](#)



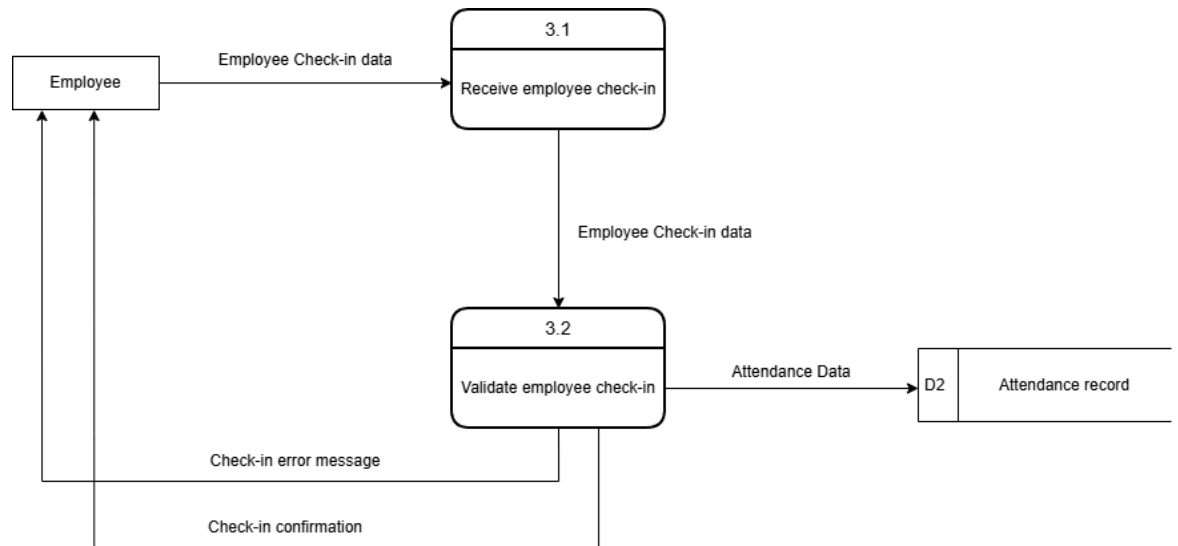
4.2.2 DFD Level 2

4.2.2.1 Sprint 1

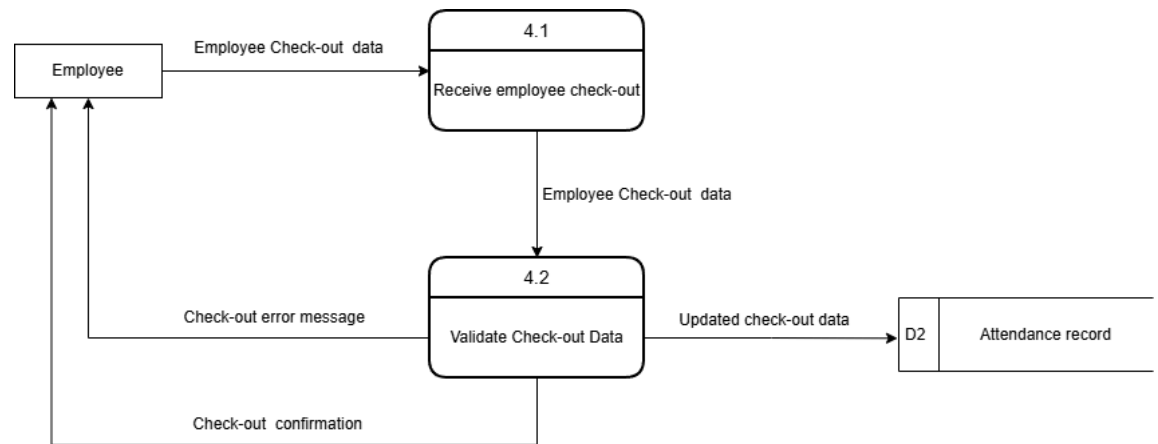


4.2.2.2 Sprint 2

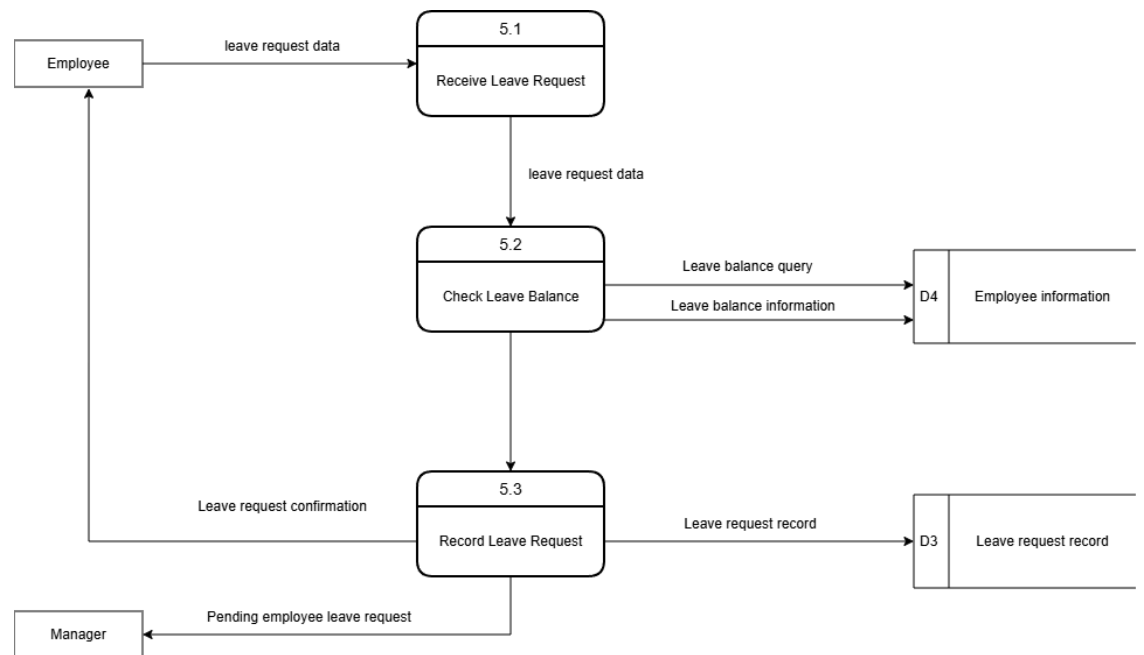
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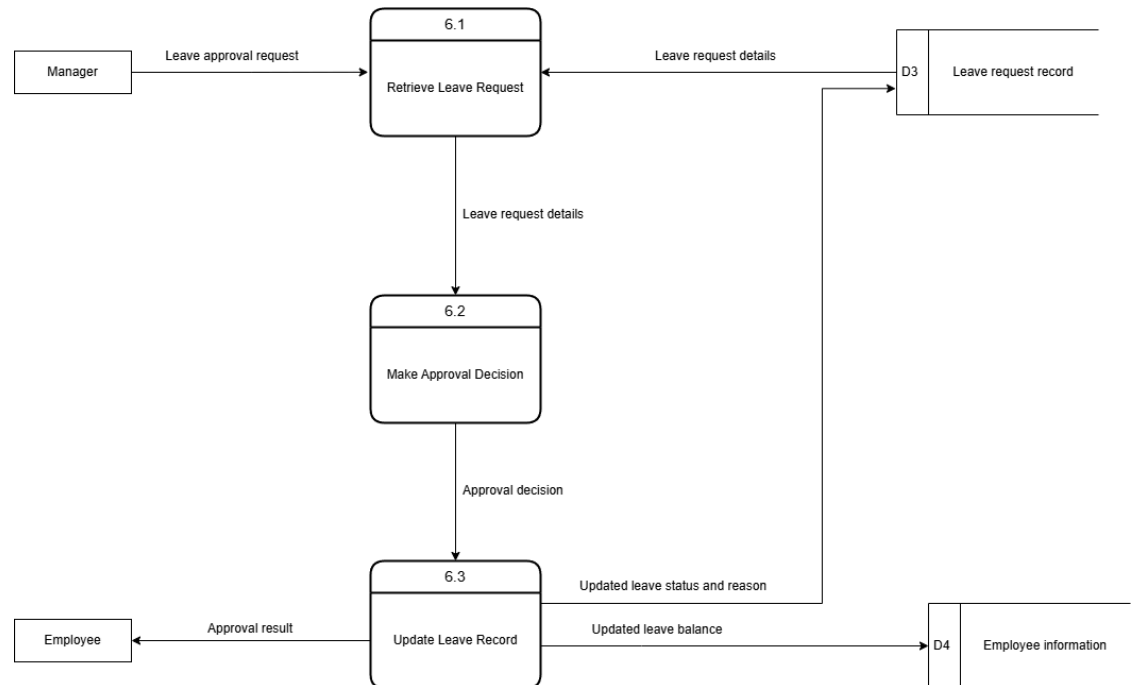
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- 2.3

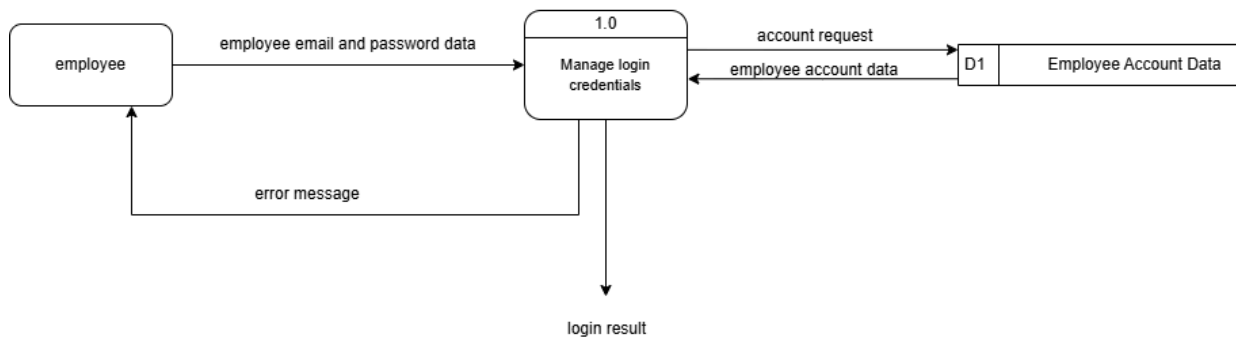


- 2.4



4.3 Process Description

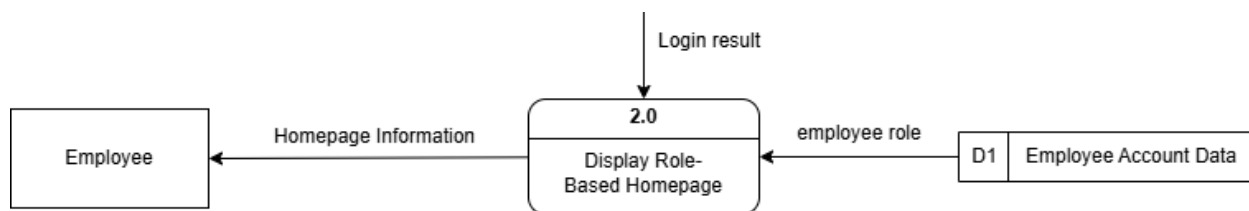
4.3.1 Sprint 1



4.3.1.1 Process Description 1.0 Manage Login Credentials

ID	1.0
Name	Manage login credentials
Description	Verifies the employee's login credentials when the employee submits an

	email and password to access the system.
Input Data Flow	- Employee email and password data - Employee account data (retrieved from data store)
Output Data Flow	- Error message - Login result
Process Description	Start 1. Receive employee email and password data from the employee. 2. Retrieve employee account data from the Employee Account Data data store. 3. Verify whether the email and password match the stored account information. 3.1 If the credentials are valid, send the verified employee account data to Process 2.0 Display Role-Based Homepage. 3.2 If the credentials are invalid, send an error message to the employee and terminate the process. End
Made by	Premiga Pengtap 673040672-0

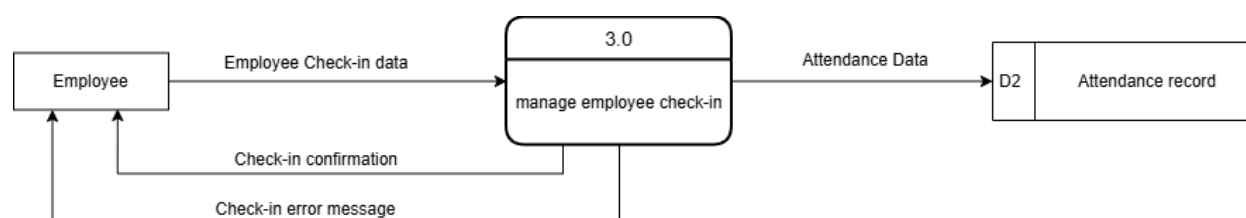


4.3.1.2 Process Description 2.0 Display Role-Based Homepage

ID	2.0
Name	Display Role-Based Homepage
Description	Displays the appropriate homepage to the employee based on their assigned role after a successful login.
Input Data Flow	- Login result
Output Data Flow	- Homepage information
Process Description	Start 1. Receive the login result from the Manage Login Credentials process. 2. Check whether the login was successful. 2.1 If login failed, terminate the process. 2.2 If login succeeded, retrieve the employee's role from the login

	result. 3. Determine the homepage based on the employee's role. 3.1 If the role is Employee, display the Employee homepage. 3.2 If the role is Manager, display the Manager homepage. 3.3 If the role is HR, display the HR homepage. 4. Send homepage information to the employee. End
Made by	Chuttiyaporn Pakdeeyut 673040145-3

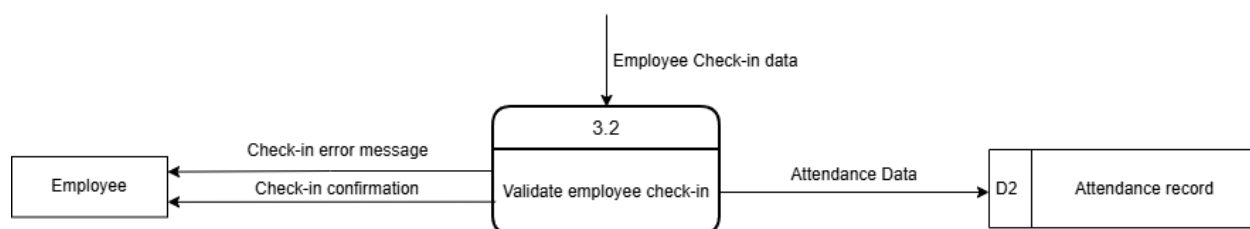
4.3.2 Sprint 2



4.3.2.1 Process Description 3.0 Record Check-in

ID	3.0
Name	Manage employee check-in
Description	Records the employee's check-in information when an employee submits a check-in request through the system. The system validates the employee information and stores the attendance data in the attendance record database.
Input Data Flow	- Employee Check-in data
Output Data Flow	- Attendance data - Check-in confirmation - Check-in error message
Process Description	Start 1. Receive employee check-in data from the Employee. 2. Retrieve the employee information from the Employee Information data store (D4). 3. Validate the employee information. 3.1 If the employee information is invalid, - send a check-in error message to the Employee - terminate the process 3.2 If the employee information is valid, proceed to the next step. 4. Check whether the employee has already checked in for the current day.

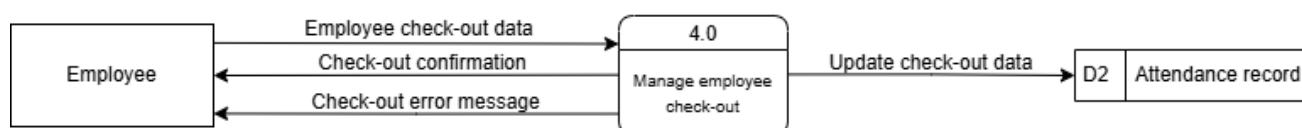
	4.1 If the employee has already checked in, - send a check-in error message to the Employee - terminate the process 4.2 If the employee has not checked in, continue. 5. Record the attendance data in the Attendance Record data store (D2). 6. Send a check-in confirmation to the Employee. End
Made by	Bunyaphop Promburom 673040462-1



4.3.2.2 Process Description 3.2 Validate Employee Check-in

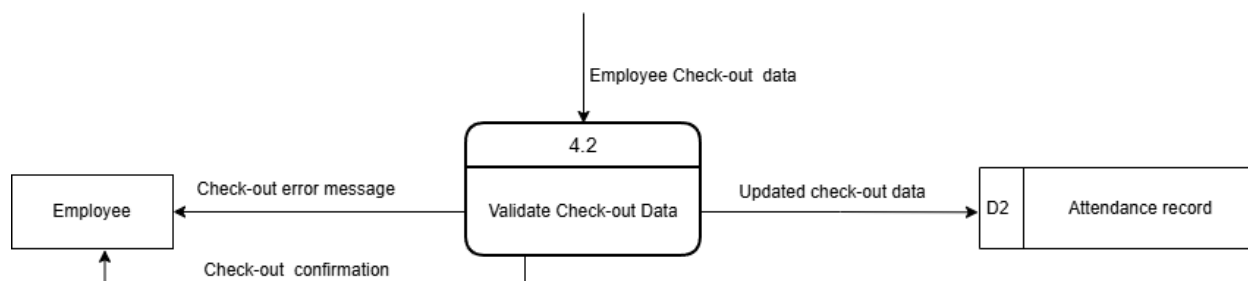
ID	3.2
Name	Validate Employee Check-In
Description	Validates the employee check-in request and records attendance data if valid.
Input Data Flow	- Employee Check-in data
Output Data Flow	- Attendance data - Check-in confirmation - Check-in error message
Process Description	Start 1. Receive employee check-in data from Process 3.1. 2. Retrieve employee information from the Employee Information data store (D4). 3. Validate the employee's eligibility to check in. 3.1 If employee information is invalid, - send a check-in error message to the Employee - terminate the process 3.2 If valid, continue. 4. Check whether the employee has already checked in for the current day. 4.1 If already checked in, - send a check-in error message to the Employee - terminate the process

	4.2 If not checked in, continue. 5. Record attendance data in the Attendance Record data store (D2). 6. Send check-in confirmation to the Employee. End
Made by	Bunyaphop Promburom 673040462-1



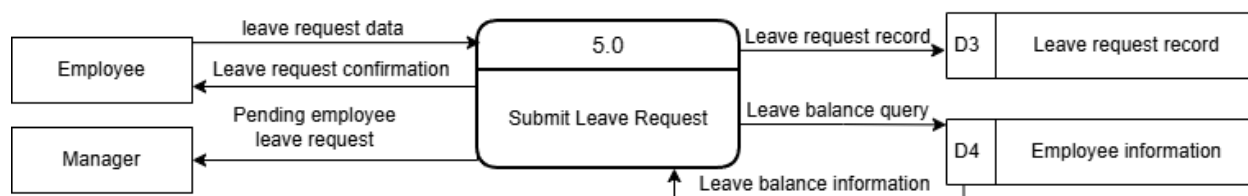
4.3.2.3 Process Description 4.0 Manage Employee Check-out

ID	4.0
Name	Manage Employee Check-Out
Description	Records the employee's check-out time, validates the request, and updates the attendance record in the system. Send confirmation or error message back to the employee.
Input Data Flow	- Employee Check-out data
Output Data Flow	- Check-out confirmation - Check-out error message - Updated check-out data
Process Description	Start 1. Receive Employee Check-out data from the Employee. 2. Validate the check-out data. 2.1 If the data is invalid, send a Check-out error message to the Employee and terminate the process. 2.2 If the data is valid, proceed to the next step. 3. Record the check-out time for the employee. 4. Send Updated check-out data to D2 - Attendance record. 5. Send Check-out confirmation to the Employee. End
Made by	Patrawadee Wonghan 673040464-7



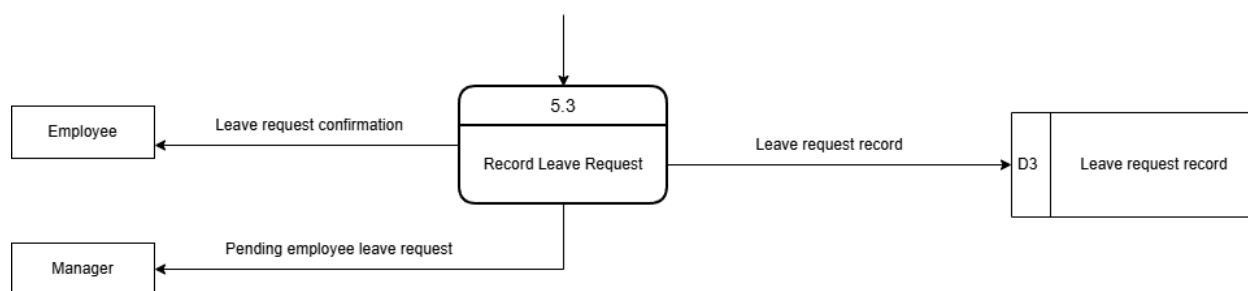
4.3.2.4 Process Description 4.2 Validate Check-Out Data

ID	4.2
Name	Validate Check-Out Data
Description	Validates the employee's check-out request, records the check-out time in the attendance system, and returns confirmation or an error message to the employee.
Input Data Flow	- Employee Check-out data
Output Data Flow	- Updated check-out data - Check-out confirmation - Check-out error message
Process Description	Start 1. Receive employee check-out data from Process 4.1 Receive Employee Check-Out. 2. Retrieve the employee's attendance record for the current day from the Attendance Record data store (D2). 3. Validate the check-out request. 3.1 If no check-in record exists for the current day, - send a check-out error message to the Employee - terminate the process 3.2 If the check-out data is invalid (e.g., incorrect format or missing data), - send a check-out error message to the Employee - terminate the process 3.3 If the request is valid, continue to the next step. 4. Record the check-out time in the End
Made by	Bunyaphop Promburom 673040462-1



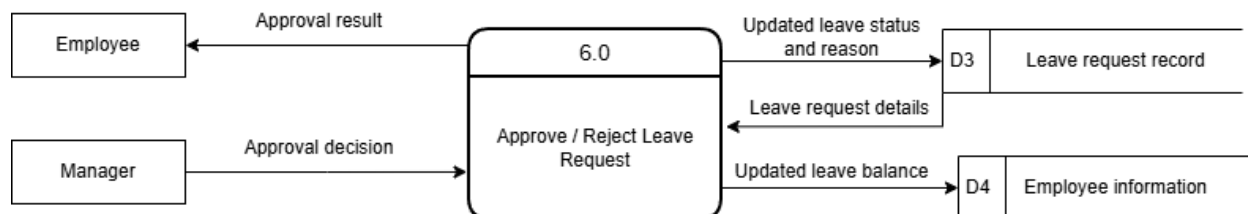
4.3.2.5 Process Description 5.0 Submit Leave Request

ID	5.0
Name	Submit Leave Request
Description	Allows an employee to submit a leave request to the system. The system records the request, checks the employee's leave balance, and sends the request to the manager for review.
Input Data Flow	- Leave request data - Leave balance information
Output Data Flow	- Leave request confirmation - Pending employee leave request - Leave request record - Leave balance query
Process Description	Start 1. Receive leave request data from the Employee. 2. Retrieve the employee's leave balance information from the Employee Information data store (D4). 3. Check whether the employee has sufficient leave balance. 3.1 If the leave balance is insufficient, terminate the process. 3.2 If the leave balance is sufficient, continue the process. 4. Store the leave request data in the Leave Request Record data store (D3). 5. Send a leave request confirmation to the Employee. 6. Send the pending employee leave request to the Manager for approval. End
Made by	Praweena Bunkham 673040463-9



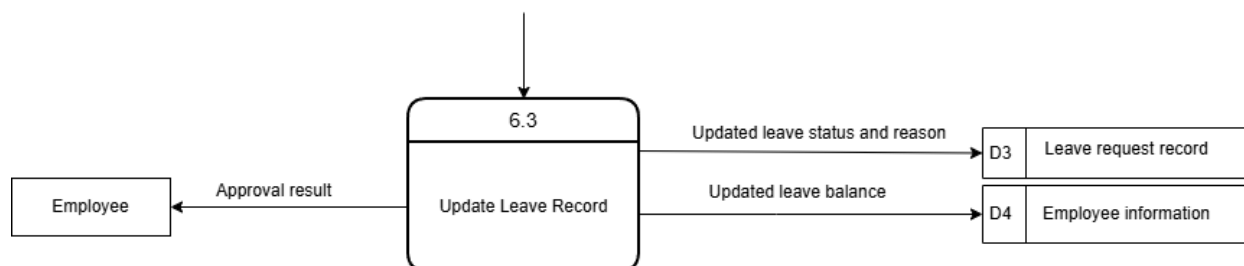
4.3.2.6 Process Description 5.3 Record Leave Request

ID	5.3
Name	Record Leave Request
Description	Records the employee's leave request in the system after verifying sufficient leave balance and forwards the request for approval.
Input Data Flow	- Leave request data - Leave balance information
Output Data Flow	- Leave request record - Pending employee leave request - Leave request confirmation
Process Description	Start 1. Receive leave request data and leave balance information from Process 5.2 Check Leave Balance. 2. Verify that the leave balance is sufficient for the requested leave period. 2.1 If the leave balance is insufficient, - send a leave request error message to the Employee - terminate the process 2.2 If the leave balance is sufficient, continue to the next step. 3. Record the leave request data in the Leave Request Record data store (D3). 4. Set the leave request status to Pending. 5. Send leave request confirmation to the Employee. 6. Send the pending employee leave request to the Manager for approval. End
Made by	Bunyaphop Promburom 673040462-1



4.3.2.7 Process Description 6.0 Approve / Reject Leave Request

ID	6.0
Name	Approve / Reject Leave Request
Description	Allows a manager to review a leave request submitted by an employee and decide whether to approve or reject it. The system then updates the leave request record and the employee's leave balance accordingly.
Input Data Flow	- Approval decision - Leave request details
Output Data Flow	- Approval result - Updated leave status - Updated leave balance
Process Description	Start 1.Receive the approval decision from the Manager. 2.Retrieve the leave request details from the Leave Request Record (D3). 3.Check the manager's decision. 4.Decision: Approved? 4.1 If the request is approved: Update the leave request status to Approved in the Leave Request Record (D3). 4.2 If the request is rejected: Update the leave request status to Rejected in the Leave Request Record (D3). 5.If the request is approved, update the employee's leave balance in the Employee Information data store (D4). 6.Send the approval result to the Employee. End
Made by	Nanticha Promwong 673040461-3



4.3.2.8 Process Description 6.3 Update Leave Record

ID	6.3
Name	Update Leave Record
Description	Updates the leave request status based on the manager's approval decision, adjusts the employee's leave balance, and notifies the employee of the approval result.
Input Data Flow	- Approval decision - Leave request details
Output Data Flow	- Updated leave status and reason - Updated leave balance - Approval result
Process Description	Start 1. Receive the approval decision from Process 6.2 Make Approval Decision. 2. Retrieve the corresponding leave request details. 3. Update the leave request status. 3.1 If the decision is Approved, - set status to "Approved" 3.2 If the decision is Rejected, - set status to "Rejected" - record the rejection reason 4. Update the leave request record in the Leave Request Record data store (D3). 5. If the leave request is approved, update the employee's leave balance in the Employee Information data store (D4). 6. Send the approval result to the Employee. End
Made by	Bunyaphop Promburom 673040462-1

4.4 Data Dictionary

4.4.1 External Entity Description (Bunyaphop Promburom 673040462-1)

Name	Description	Input Data Flow	Output Data Flow
Employee	A system user (Officer) who records attendance, submits leave requests, and views personal information.	- Email and password data - Check-in data - Check-out data - Leave request data	- Homepage information - Error messages - Check-in confirmation - Check-out confirmation - Leave request confirmation - Approval result
Manager	A supervisor responsible for reviewing attendance and approving or rejecting leave requests of team members.	- Leave approval request - Approval decision	- Pending employee leave requests - Leave request details

4.4.2 Data Flow Description and Data Structure (Bunyaphop Promburom 673040462-1)

Name	Description	Source	Destination	Data Structure
Employee Email and Password Data	Details of login credentials submitted by the employee	Employee	Process 1.0 Manage Login Credentials	Email + Password
		Process 1.0 Manage Login Credentials	D1 Employee Account Data	
Employee Account Data	Stored account information of employees	D1 Employee Account Data	Process 1.0 Manage Login Credentials	Employee_ID + Email + Password + Role
		Process 1.0 Manage Login Credentials	Employee	

Employee Check-In Data	Information submitted to record check-in time	Employee	Process 3.0 Manage Employee Check-In	Employee_ID + Date + Check_in_time + Location
		Process 3.0 Manage Employee Check-In	D1 Attendance Record	
			Employee	
Employee Check-Out Data	Information submitted to record check-out time	Employee	Process 4.0 Manage Employee Check-Out	Employee_ID + Date + Check_out_time + Location
		Process 4.0 Manage Employee Check-Out	D2 Attendance Record	
			Employee	
Leave Request Data	Details of leave request submitted by employee	Employee	Process 5.0 Submit Employee Leave Request	Employee_ID + Leave_type + Start_date + End_date + Reason
		Process 5.0 Submit Employee Leave Request	D3 Leave Request Record	
			Manager	
			Employee	
Leave Balance Query	Request to retrieve employee leave balance	Process 5.0 Submit Employee Leave Request	D4 Employee Information	Employee_ID
		Process 6.0 Process Leave Approval		
Leave Balance Information	Available leave balance returned from database	D4 Employee Information	Process 5.0 Submit Employee Leave Request	Employee_ID + Leave_balance
Leave Approval Request	Approval or rejection decision submitted by manager	Manager	Process 6.0 Process Leave Approval	Leave_id + Manager_id + Approval_status + Comment
		Process 6.0 Process Leave	D3 Leave Request Record	

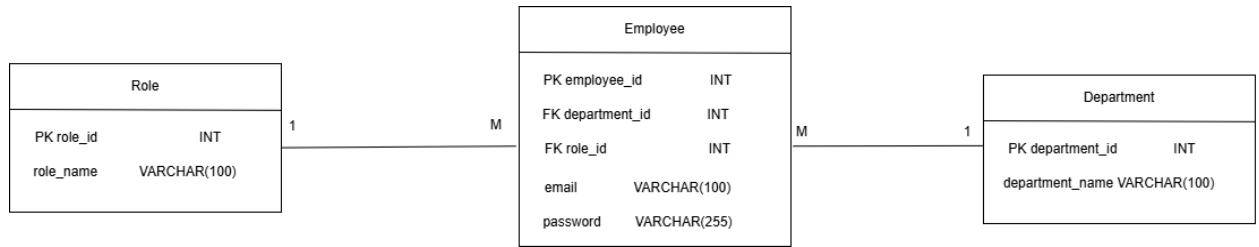
		Approval	D4 Employee Information	
			Employee	
Leave Request Details	Detailed leave request retrieved for decision making	D3 Leave Request Record	Process 6.0 Process Leave Approval	Leave_ID + Employee_ID + Leave_type + Dates + Reason + Status

4.4.3 Data Store Description

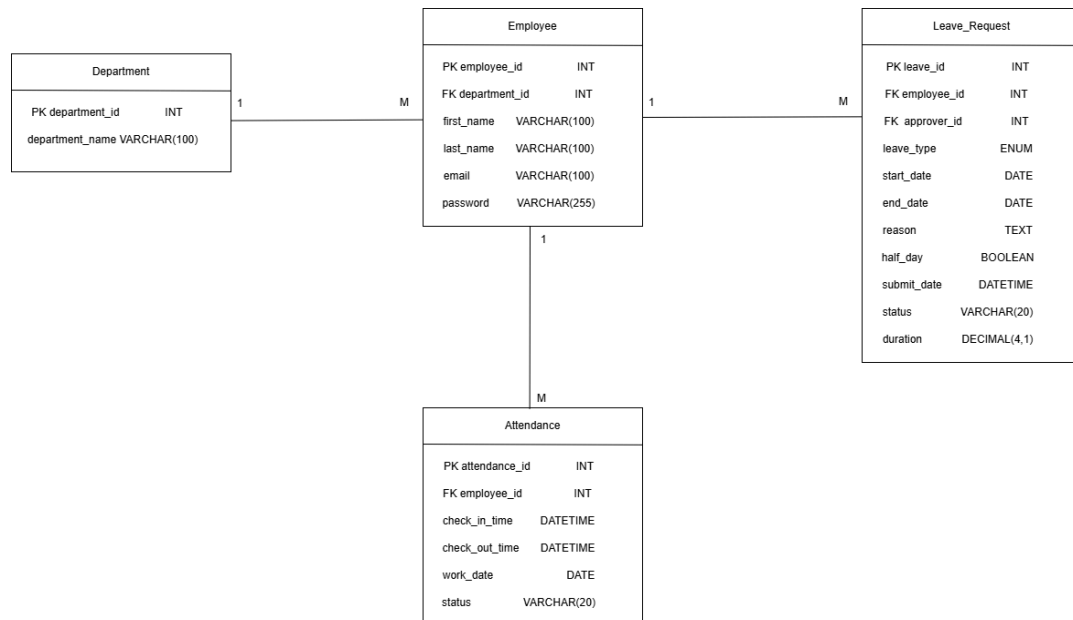
ID	Data Store Name	Description	Data Structure	Made by
D1	Employee account Data	Stores employee account data for authentication and role identification.	<u>employee_id</u> + username + password + first_name + last_name + email + role + account_status + last_login	Patrawadee Wonghan 673040464-7
D2	Attendance record	Stores employee attendance check-in and check-out records.	<u>attendance_id</u> + employee_id + date + check_in_time + check_out_time + location + status	Chuttiyaporn Pakdeeyut 673040145-3
D3	Leave request record	Stores employee leave request data and leave request status.	<u>leave_request_id</u> + employee_id + leave_type + start_date + end_date + leave_request_data + leave_request_status	Praweena Bunkham 673040463-9
D4	Employee information	Stores employee information.	<u>employee_id</u> + first_name + last_name + email + phone_number + position + department + employment_status + leave_balance + supervisor_id	Bunyaphop Promburom 673040462-1

4.5 ER Diagram

4.5.1 Sprint 1 [DFD ER Sprint 1 Draw.io](#)



4.5.2 Sprint 2 [DFD ER Sprint 2 Draw.io](#)



Chapter 5 - Sprint Implementation

5.1 Sprint 1

5.1.1 Sprint Goal

The goal of Sprint 1 was to establish the foundation of the system by implementing secure user access and basic system navigation. This sprint focused on enabling users to log into the website safely and access the system according to their roles.

By the end of this sprint, users should be able to authenticate themselves, recover their passwords if necessary, and access their respective dashboards. These features are essential because they provide the entry point to all other system functionalities.

5.1.2 Selected Backlog Items

The backlog items selected for Sprint 1 focused on authentication and access control features. These items were prioritized because they are required before implementing operational functions such as attendance recording and leave management. The selected backlog items include:

- Secure Login System
- Role-Based Access Control
- Role-Based Homepage Dashboard

These features ensure that the system can identify users correctly and provide appropriate access permissions.

5.1.3 Process Description

Login Process

1. The user opens the system login page.
2. The user enters their email and password.
3. The system receives the login credentials.
4. The system validates the credentials against the employee account database.
5. If the credentials are valid, the system identifies the user's role (Officer, Manager, HR).
6. The system retrieves the user's profile information.
7. The system redirects the user to the corresponding homepage.
8. If the credentials are invalid, the system displays an error message and denies access.

Failed Login Handling Process

1. The user submits invalid credentials.
2. The system detects that the credentials do not match any account.
3. The system prevents access to the system.
4. The system displays an error notification indicating invalid username or password.
5. The user remains on the login page and may attempt to log in again.

Homepage Display Process

1. After successful login, the system determines the user's role.
2. The system retrieves relevant data for that role.
3. The system loads the corresponding homepage interface:
 - Officer Homepage
 - Manager Homepage
 - HR Homepage
4. The system displays available functions and information for the user.
5. The user can access system features permitted by their role.

5.1.4 Increment

The increment produced at the end of Sprint 1 is a functional authentication module that allows users to access the website securely. Users can log into the system using their credentials and are directed to role-specific dashboards based on their permissions.

The system also supports password recovery through the forgot password feature, enabling users to regain access to their accounts if needed. The dashboards provide a basic interface for navigating the system, although operational features such as attendance recording are implemented in later sprints.

This increment establishes the core infrastructure of the website and ensures that users can access the system safely before additional functionalities are developed.

5.2 Sprint 2

5.2.1 Sprint Goal

The goal of Sprint 2 was to implement the core operational features of the website, specifically attendance management and leave processing. This sprint focused on enabling employees to record their working hours and submit leave requests, as well as allowing managers or HR personnel to review and approve those requests.

By the end of this sprint, the system should support daily attendance recording, leave request submission, approval workflows, and viewing of attendance information. These features represent the main functions that users interact with during regular system usage.

5.2.2 Selected Backlog Items

The backlog items selected for Sprint 2 focused on the primary business functions of the system. These items were prioritized after the authentication features were completed in Sprint 1. The selected backlog items include:

- Check-in Function
- Check-out Function
- Attendance History Viewing
- Leave Request Submission
- Leave Approval by Manager or HR
- Leave Status Tracking

These features enable the system to support day-to-day attendance monitoring and leave management within the organization.

5.2.3 Process Description

The process descriptions explain the system workflow for the functions implemented in Sprint 2.

Check-in Process

1. The user logs into the system.
2. The user selects the Check-In function.
3. The system verifies that the user has not already checked in for the current day.
4. The system checks the current time and location (if applicable).
5. If all conditions are valid, the system records the check-in time.
6. The attendance record is stored in the database.
7. The system displays confirmation of successful check-in.

Check-out Process

1. The user logs into the system.
2. The user selects the Check-Out function.
3. The system verifies that the user has already checked in for the current day.
4. The system checks the current time and location (if applicable).
5. If conditions are valid, the system records the check-out time.

6. The attendance record is updated in the database.
7. The system displays confirmation of successful check-out.

Leave Request Process

1. The user selects the Leave Request function.
2. The system displays the leave request form.
3. The user selects the leave type.
4. The user selects the start date and end date.
5. The user enters the reason for leaving.
6. The system validates the input data and checks the remaining leave balance.
7. If the request is valid, the system records the leave request in the database.
8. The leave request status is set to Pending.
9. The system sends the request to the manager or HR for approval.
10. The system displays confirmation of successful submission.

5.2.4 Increment

The increment produced at the end of Sprint 2 is a functional attendance and leave management module that supports daily organizational operations.

Employees can record their attendance through check-in and check-out features, submit leave requests, and view their attendance history. Managers or HR personnel can review and approve leave requests, ensuring that leave processes are handled efficiently.

This increment represents the core functionality of the website, enabling real-world use within the organization.

5.3 Test Scenarios

User story ID	Test scenario ID	Test scenario	Number of test cases	Priority
US0001	TS0001	Verify that the system displays an error message when the user enters an incorrect username or password	3	High
US0001	TS0002	Verify that the user can log in successfully with HR credentials and is redirected to the HR homepage	1	High
US0001	TS0003	Verify that the user can log in successfully with Worker credentials and is redirected to the Worker homepage	1	High
US0001	TS0004	Verify that the user can log in successfully with Manager credentials and is redirected to the Manager homepage	1	High
US003	TS0005	Verify whether employee can check in successfully	1	High
US003	TS0006	Verify whether employee can't check in successfully	3	High
US003	TS0007	Verify whether employee can check out successfully	1	High
US003	TS0008	Verify whether employee can't check out successfully	4	High
US004	TS0009	Verify whether user can submit a leave request with valid inputs	1	High
US004	TS0010	Verify whether the system handles incomplete inputs on the leave request page	5	High
US004	TS0011	Verify whether the system saves and displays the submitted leave request correctly after submission	3	High

5.4 Test case

Test Scenario	Test Case ID	Test Case	Pre-Condition	Test Step	Test Data	Expect Result	Post Condition	Actual Result	Status (Pass/Fail)
TS0001: Verify that the system displays an error message when the user enters an incorrect username or password	TC01	Verify that entering incorrect username shows error	User has a valid account in the system and is on the Login Page	1.Open the web 2.System displays the Login Page 3.Click on the Username / Email field 4.Enter an incorrect Username / Email 5.Click on the Password field 6.Enter a Password 7.Click the Login button	1.Username / Email = (Incorrect) 2.Password = (Correct)	1.System processes login request 2.System displays error message: "Invalid username" 3.User remains on Login Page	User stays on Login Page and cannot access the system	User stays on Login Page but cannot displays error message	Fail
TS0001: Verify that the system displays an error message when the user enters an incorrect username or password	TC02	Verify that entering incorrect password only shows error	User has a valid account in the system and is on the Login Page	1.Open the web 2.System displays the Login Page 3.Click on the Username / Email field 4.Enter a Username / Email 5.Click on the Password field 6.Enter an incorrect Password 7.Click the Login button	1.Username / Email = (Correct) 2.Password = (Incorrect)	1.System does not allow login 2.System displays an error message: "Invalid password" 3.User remains on the Login Page	User remains on Login Page and cannot access any system features	User stays on Login Page but cannot displays error message	Fail
TS0001: Verify that the system displays an error message when the user enters an incorrect username or password	TC03	Verify that leaving username and password empty shows error	User has a valid account in the system and is on the Login Page	1.Open the web 2.System displays the Login Page 3.Leave the Username / Email field empty. 4.Click the Login button	1.Username / Email = (Empty) 2.Password = (Empty)	1.System does not allow login. 2.System displays an error message indicating that Username and Password are required. 3.User remains on the Login Page.	User stays on Login Page and cannot access the system	User stays on Login Page but cannot displays error message	Fail
TS0002:	TC01	Verify that	1. HR	1.Open a web	1.Username	1.Login is	User is	The system	Pass

Verify that the user can log in successfully with HR credentials and is redirected to the HR homepage		the user can log in successfully with HR credentials and is redirected to the HR homepage	accounts already exist in the system. 2.User is on the Login Page. 3.System is running normally and connected to the database.	2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter the correct password 7.Click the Login button 8.System validates the credentials 9.System identifies the user role as HR 10.System redirects the user to the HR Homepage	/ Email = (Correct) 2.Password = (Correct) 3.Role = HR	successful 2.System redirects the user to the HR Homepage 3. HR menu or dashboard is displayed 4.User can access features that are available for the HR role only	logged into the system and user on the HR Homepage	allows the user to log in successfully and redirects the user to the HR Homepage. The HR dashboard is displayed correctly and the user can access the HR home page.	
TS0003: Verify that the user can log in successfully with Worker credentials and is redirected to the Worker homepage	TC01	Verify that the user can log in successfully with Worker credentials and is redirected to the Worker homepage	1.Worker account already exists in the system. 2.User is on the Login Page. 3.System is running normally and connected to the database.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter the correct password 7.Click the Login button 8.System validates the credentials 9.System identifies the user role as Worker 10.System redirects the user to the Worker Homepage	1.Username / Email = (Correct) 2.Password = (Correct) 3.Role = Worker	1.Login is successful 2.System redirects the user to the Worker Homepage 3.Worker menu or dashboard is displayed 4.User can access features that are available for the Worker role only	User is logged into the system and user on the Worker Homepage	The system allows the user to log in successfully and redirects the user to the Worker Homepage. The Worker dashboard is displayed correctly and the user can access the Worker home page.	Pass
TS0004: Verify that the user can log in successfully with Manager credentials and is redirected to the	TC01	Verify that the user can log in successfully with Manager credentials and is redirected to the Manager	1.Manager account already exists in the system. 2.User is on the Login Page. 3.System is running normally	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter the correct	1.Username / Email = (Correct) 2.Password = (Correct) 3.Role = Manager	1.Login is successful 2.System redirects the user to the Manager Homepage 3.Manager menu or dashboard is displayed	User is logged into the system and user on the Manager Homepage	The system allows the user to log in successfully and redirects the user to the Manager Homepage.	Pass

Manager homepage		homepage	and connected to the database.	password 7.Click the Login button 8.System validates the credentials 9.System identifies the user role as Manager 10.System redirects the user to the Manager Homepage		4.User can access features that are available for the Manager role only		The Manager dashboard is displayed correctly and the user can access the Manager home page.	
TS0005 : Verify whether employee can check in successfully	TC01	Verify employee can check in successfully when all conditions are valid	1.Employee account exists in the system. 2.Employee is registered in the attendance system. 3.Employee is on the Home Page / Attendance Page. 4.Employee has not checked in for the current day.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Locate the Check In button on the right panel. 10.Click the Check In button. 11.System records the employee's check-in time. 12.System updates the attendance status.	1.Username / Email = (Correct) 2.Password = (Correct) 3.Date = Current working day 4.Check-in time = 08:55 AM	The system records the employee check-in time successfully and displays the check-in status as "On time" when the employee checks in within the scheduled time.	The employee attendance record for the current day is saved in the system database.	The system successfully records the check-in time and displays the check-in status as "On time" in the schedule section.	Pass
TS0006 : Verify whether employee can't check in successfully	TC01	Verify system shows error when employee checks in outside allowed location	1.Employee account exists in the system. 2.Employee is registered in the attendance system. 3.System requires location	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password	1.Username / Email = (Correct) 2.Password = (Correct) 3.Current Location = Outside company area	1.System does not allow check-in. 2.System displays an error message 3.Check-in time is not recorded in the system. 4.User	No attendance record is created for the current check-in attempt.	The system prevents check-in and displays an error message indicating that the employee is outside the allowed	Pass

			verification (GPS / allowed location) for check-in. 4.Employee is currently outside the allowed company location. 5.Employee is on the Employee Homepage.	7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Locate the Check In button on the dashboard. 10.Ensure the device location is inside the allowed company location. 11.Click the Check In button. 12.System verifies the user location.		remains on the Employee Homepage.		location.	
TS0006 : Verify whether employee can't check in successfully	TC02	Verify system shows error when employee checks in before the allowed check-in time (change the button to be disabled)	1.Employee account exists in the system. 2.Employee is registered in the attendance system. 3.System has a defined allowed check-in time. 4.Employee has not checked in yet for the current day. 5.Employee is on the Employee Homepage.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Check the current time on the device. 10.Locate the Check In button on the dashboard. 11.Click the Check In button. 12.System verifies the current time against the allowed check-in schedule.	1.Username / Email = (Correct) 2.Password = (Correct) 3.Allowed Check-in Time = 08:30 AM 4.Attempt Check-in Time = 07:45 AM	1.System does not allow check-in. 2.The Check In button is disabled and cannot be clicked. 3.Check-in time is not recorded in the attendance system. 4.User remains on the Employee Homepage.	No attendance record is created for this check-in attempt.	The system prevents check-in before the allowed time by disabling the Check In button, so the employee cannot click the button.	Pass
TS0006 : Verify whether employee can't check in successfully	TC03	Verify system shows error when employee tries to check in	1.Employee account exists in the system. 2.Employee is registered in the	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid	1.Username / Email = (Correct) 2.Password = (Correct) 3.First Check-in	The system does not allow the employee to check in again after already checking in.	Only the first check-in record remains stored in the system	After the employee checks in successfully, the Check In button is	Pass

y		again after already checking in (change the button to be disabled)	attendance system. 3.Employee has already checked in successfully for the current day. 4.Employee is on the Employee Homepage.	username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Click the Check In button to perform the first check-in. 10.System records the check-in time successfully. 11.Attempt to click the Check In button again.	Time = 08:45 AM 4.Second Attempt Time = 08:50 AM	The Check In button becomes disabled and cannot be clicked.	for the current day.	disabled and cannot be clicked again, preventing multiple check-ins.	
TS0007: Verify whether employee can check out successfully	TC01	Verify that the employee can check out successfully	1.Employee account exists in the system. 2.Employee has already checked in for the current day. 3.Employee is logged into the system. 4.Employee is on the Employee Homepage / Attendance page.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Verify that the employee has already checked in for the day. 10.Locate the Check Out button on the attendance panel. 11.Click the Check Out button. 12.System records the employee's check-out time.	1.Username / Email = (Correct) 2.Password = (Correct) 3.Check-in Time = 08:45 AM 4.Check-out Time = 05:00 PM	1.System successfully records the check-out time. 2.System displays the recorded check-out time on the attendance or schedule panel. 3.Attendance status for the day is updated.	Employee attendance record for the current day includes both check-in and check-out time in the system.	The system records the check-out time successfully and displays it in the attendance schedule.	Pass
TS0008:Ve	TC01	Verify that	1.Employee	1.Open a web	1.Username	1.System	No	The system	Pass

Verify whether employee can't check out successfully		the system does not allow the user to check out when the user is outside the allowed location	account exists in the system. 2.Employee has already checked in for the current day. 3.System requires location verification (GPS / allowed location) for check-out. 4.Employee is currently outside the allowed company location. 5.Employee is logged in and on the Employee Homepage.	2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Verify that the employee has already checked in for the current day. 10.Move the device to a location outside the allowed company area. 11.Click the Check Out button. 12.System verifies the user's location.	/ Email = (Correct) 2.Password = (Correct) 3.Current Location = Outside company area	does not allow check-out. 2.System displays an error message. 3.Check-out time is not recorded in the system.	check-out record is created for this attempt.	prevents check-out because the employee is outside the allowed location and does not record the check-out time.	
TS0008: Verify whether employee can't check out successfully	TC02	Verify that the system does not allow the user to check out without checking in first (button is disabled)	1.Employee account exists in the system. 2.Employee has not checked in for the current day. 3.System requires employees to check in before check-out. 4.Employee is logged in and on the Employee Homepage.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Verify that the employee has not checked in for the current day. 10.Locate the Check Out button	1.Username / Email = (Correct) 2.Password = (Correct)	1.System does not allow check-out. 2.The Check Out button is disabled or cannot be clicked. 3.Check-out time is not recorded in the system.	No check-out record is created for this attempt.	The system prevents check-out because the employee has not checked in for the current day and does not record any check-out time.	Pass

				on the Employee Homepage. 11.Attempt to click the Check Out button.					
TS0008: Verify whether employee can't check out successfully	TC03	Verify that the system does not allow the user to check out again after already checking out (button is disabled)	1.Employee account exists in the system. 2.Employee has already checked in and checked out for the current day. 3.System allows only one check-out record per day. 4.Employee is logged in and on the Employee Homepage.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Verify that the employee has already checked in and checked out for the current day. 10.Locate the Check Out button on the Employee Homepage. 11.Attempt to click the Check Out button again.	1.Username / Email = (Correct) 2.Password = (Correct)	1.System does not allow another check-out attempt. 2.The Check Out button is disabled or unavailable. 3. No additional check-out time is recorded in the system.	The existing check-out record remains unchanged .	The system prevents duplicate check-out attempts because the employee has already checked out for the current day.	Pass
TS0008: Verify whether employee can't check out successfully	TC04	Verify that the system does not allow the user to check out before the allowed check-out time (button is disabled)	1.Employee account exists in the system. 2.Employee has already checked in for the current day. 3.System has a defined allowed check-out time. 4.Current	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects	1.Username / Email = (Correct) 2.Password = (Correct) 3.Current Time = Before allowed check-out time	1.System does not allow check-out. 2.System displays a message indicating that check-out is not allowed yet. 3.Check-out time is not recorded in the system.	No check-out record is created for this attempt.	The system prevents early check-out according to company policy and does not record the check-out time.	Pass

			time is before the allowed check-out time. 5.Employee is logged in and on the Employee Homepage.	the user to the Employee Homepage. 9.Verify that the employee has already checked in for the current day. 10.Verify that the current time is before the allowed check-out time. 11.Locate the Check Out button on the Employee Homepage. 12.Attempt to click the Check Out button.					
TS0009: Verify whether user can submit a leave request with valid inputs	TC01	Verify that user can submit leave request successfully when all required fields are filled with valid inputs	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Navigate to the Leave Request Page. 10.Click the Request button. 11.Select a Leave Type. 12.Enter a valid Start Date. 13.Enter a valid End Date. 14.Enter a valid reason for leaving. 15.Enter a valid Duration for leave. 16.Click the Submit button.	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 10/03/2026 5.End Date = 11/03/2026 6.Reason = Medical appointment 7.Duration = 2 Days	1.System successfully saves the leave request. 2.System displays a confirmation message such as "Leave request submitted". 3.The submitted leave request appears in the Leave Request History 4.The leave details (type, date, reason, status) are displayed correctly.	The leave request is successfully recorded in the system and appears in the leave request history.	The system successfully records the leave request and displays the confirmation message.	Pass

TS0010: Verify whether the system handles incomplete inputs on the leave request page	TC01	Verify system shows error when Leave Type is not selected	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Navigate to the Leave Request Page. 10.Click the Request button. 11.Leave the Leave Type field empty. 12.Enter a valid Start Date. 13.Enter a valid End Date. 14.Enter a valid reason for leaving. 15.Enter a valid Duration for leave. 16.Click the Submit button.	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = (Not selected) 4.Start Date = 10/03/2026 5.End Date = 11/03/2026 6.Reason = Medical appointment 7.Duration = 2 Days	1.System does not submit the leave request. 2.System displays an error message indicating that Leave Type is required. 3.The leave request is not saved in the system.	No leave request record is created.	The system prevents submission and displays an error message indicating that Leave Type is required.	Pass
TS0010: Verify whether the system handles incomplete inputs on the leave request page	TC02	Verify system shows error when Start Date is not entered	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = (Empty) 5.End Date = 11/03/2026 6.Reason = Medical appointment 7.Duration = 2 Days	1.System does not submit the leave request. 2.System displays an error message indicating that a Start Date is required. 3.The leave request is not saved in the system.	No leave request record is created.	System prevents submission and displays an error message indicating that a Start Date is required.	Pass

				Homepage. 9.Navigate to the Leave Request Page. 10.Click the Request button. 11.Select a Leave Type. 12.Leave the Start Date field empty. 13.Enter a valid End Date. 14.Enter a valid reason for leaving. 15.Enter a valid Duration for leave. 16.Click the Submit button.					
TS0010: Verify whether the system handles incomplete inputs on the leave request page	TC03	Verify system shows error when End Date is not entered	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Navigate to the Leave Request Page. 10.Click the Request button. 11.Select a Leave Type. 12.Enter a valid Start Date. 13.Leave the End Date field empty. 14.Enter a valid reason for leaving. 15.Enter a valid Duration for leave. 16.Click the Submit button.	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 10/03/2026 5.End Date = (Empty) 6.Reason = Medical appointment 7.Duration = 2 Days	1.System does not submit the leave request. 2.System displays an error message indicating that End Date is required. 3.The leave request is not saved in the system.	No leave request record is created.	The system prevents submission and displays an error message indicating that an End Date is required.	Pass

TS0010: Verify whether the system handles incomplete inputs on the leave request page	TC04	Verify system shows error when Reason is not entered	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Navigate to the Leave Request Page. 10.Click the Request button. 11.Select a Leave Type. 12.Enter a valid Start Date. 13.Enter a valid End Date. 14.Leave the Reason field empty. 15.Enter a valid Duration for leave. 16.Click the Submit button.	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 10/03/2026 5.End Date = 11/03/2026 6.Reason = (Empty) 7.Duration = 2 Days	1.System does not submit the leave request. 2.System displays an error message indicating that Reason is required. 3.The leave request is not saved in the system.	No leave request record is created.	The system prevents submission and displays an error message indicating that Reason is required.	Pass
TS0010: Verify whether the system handles incomplete inputs on the leave request page	TC05	Verify system shows error when Duration is not entered	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser. 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 10/03/2026 5.End Date = 11/03/2026 6.Reason = Medical appointment	1.System does not submit the leave request. 2.System displays an error message indicating that Duration is required. 3.The leave request is not saved in the system.	No leave request record is created.	The system prevents submission and displays an error message indicating that Duration is required.	Pass

				Employee Homepage. 9.Navigate to the Leave Request Page. 10.Click the Request button. 11.Select a Leave Type. 12.Enter a valid Start Date. 13.Enter a valid End Date. 14.Enter a valid reason for leaving. 15.Leave the Duration field empty. 16.Click the Submit button.	7.Duration = (Empty)				
TS0011: Verify whether the system saves and displays the submitted leave request correctly after submission	TC01	Verify system saves leave request after submission	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the User Homepage. 9.Click the Leave Request / Request Leave menu. 10.System displays the Leave Request Form. 11.Select the Leave Type. 12.Select the Start Date of leave. 13.Select the End Date of leave. 14.Enter the Reason for leaving in the reason field. 15.Enter Duration in the field	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 15/03/2026 5.End Date = 16/03/2026 6.Reason = Flu and need rest 7.Duration = 1	1.System successfully saves the leave request. 2.System displays a confirmation message such as "Leave request submitted". 3.The submitted leave request appears in the Leave Request History 4.The leave details (type, date, reason, status) are displayed correctly.	Leave request record is stored in the system database and displayed in the user's leave request history.	The system successfully saves the leave request and displays the submitted leave request in the Leave History page correctly.	Pass

				16.Click the Submit / Send Request button. 17.System processes and saves the leave request. 18.Navigate to Leave History / My Leave Requests page to view the submitted request.					
TS0011: Verify whether the system saves and displays the submitted leave request correctly after submission	TC02	Verify submitted leave request is displayed in leave request list	1.User account exists in the system. 2.User is logged into the system. 3.A leave request has already been submitted successfully.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the User Homepage. 9.Click the Leave Request 10.System displays the Leave Request List page. 11.Check the leave request list for the previously submitted leave request	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 15/03/2026 5.End Date = 16/03/2026 6.Reason = Flu and need rest 7.Duration = 1	1.System displays the submitted leave request in the Leave Request List. 2.Leave details such as Leave Type, Start Date, End Date, Reason, Duration and Status are shown correctly. 3.Leave request status is displayed (e.g., Pending / Submitted).	Leave requests remain stored in the system and visible in the leave request list.	The system displays the submitted leave request in the Leave Request List with correct details.	Pass
TS0011: Verify whether the system saves and displays the submitted leave request correctly after submission	TC03	Verify system displays correct leave request details	1.User account exists in the system. 2.User is logged into the system. 3.A leave request has already been submitted and saved successfully.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 15/03/2026 5.End Date = 16/03/2026 6.Reason =	1.System displays the leave request details page. 2.Leave information such as Leave Type, Start Date, End Date, Reason, and Status are displayed correctly and match the submitted	Leave request details remain stored correctly in the system database and can be viewed by the user.	The system displays the leave request details correctly and matches the submitted leave request information.	Pass

				the user to the User Homepage. 9.Click the Leave Request / My Leave menu. 10.System displays the Leave Request List page. 11.Select or click the submitted leave request record from the list. 12.System opens the Leave Request Details page. 13.Review the leave request information displayed.	Flu and need rest 7.Duration = 1	data. 3. No data is missing or incorrect.			
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5.5 Increment Meeting Requirements (Figma)

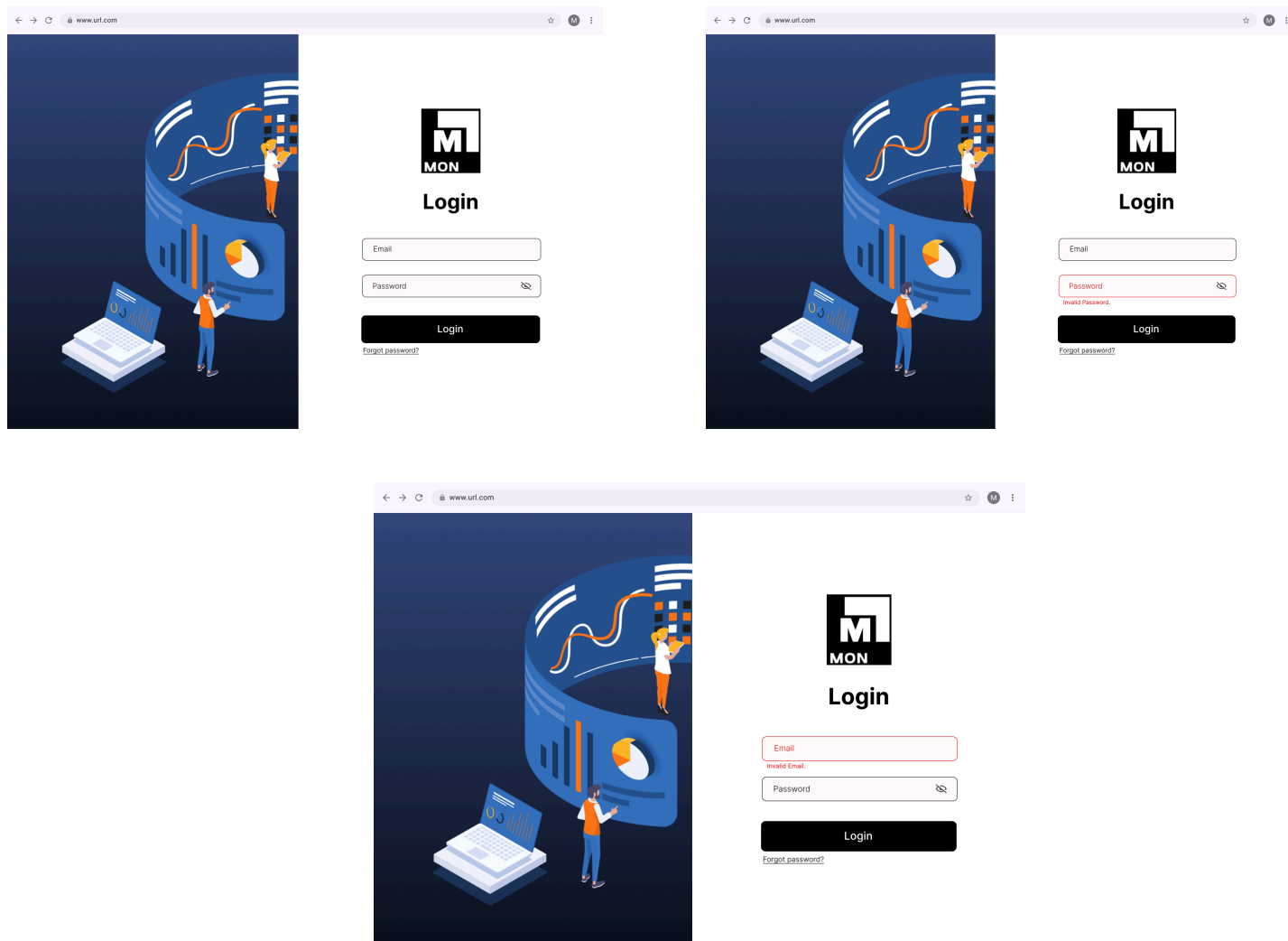
5.5.1 Sprint 1

In Sprint 1, we focused on designing the core user interface of the system in Figma. The completed designs include:

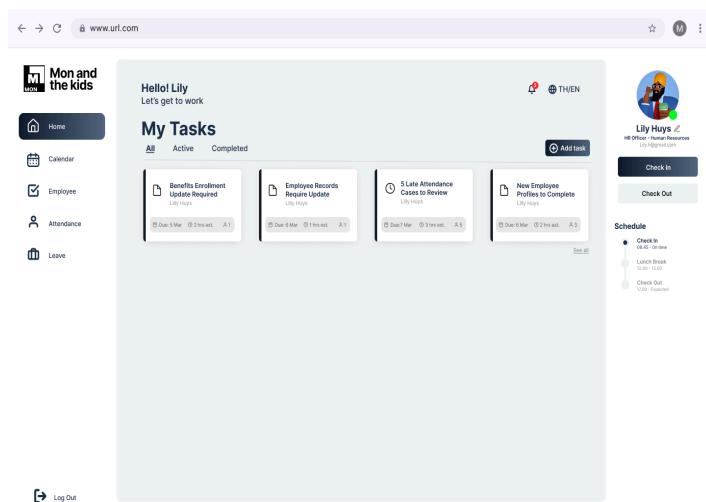
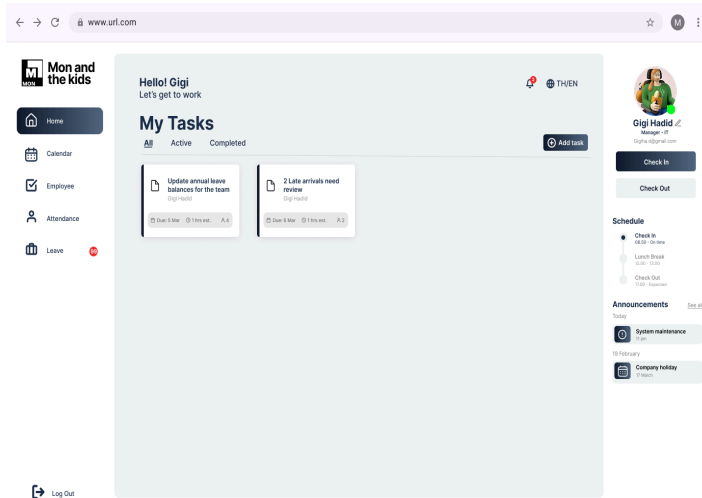
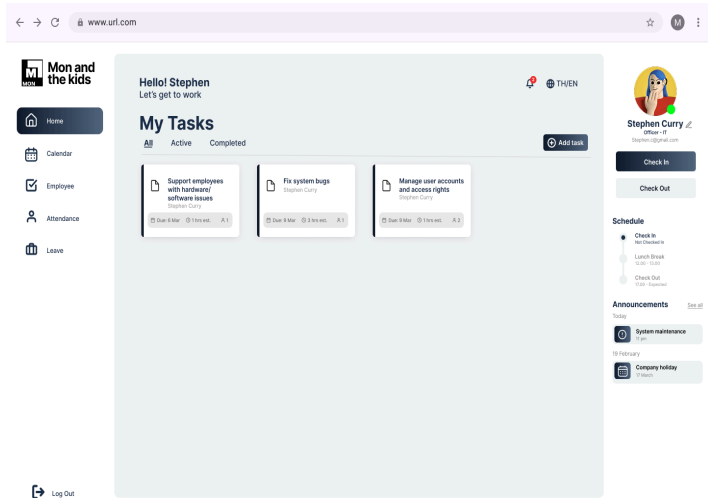
- Login page UI
- Login validation (error message for invalid password)
- Home page UI (HR, Manager, Officer roles)
- Navigation sidebar (Home, Calendar, Employee, Attendance, Leave)
- Task management dashboard (My Tasks, Active, Completed)
- Check-in / Check-out panel
- User profile display section

For functional requirements, these features support user authentication, role-based access (HR, Manager, Officer), viewing assigned tasks, managing daily work activities, and performing attendance actions such as check-in and check-out.

For non-functional requirements, these features provide a clean and modern interface, easy navigation, clear visual hierarchy, and improved usability for different user roles.



Picture 5.1 Login Page and Error Message Login Page



Picture 5.2 Based on Role Homepage(HR,Manager,Officer)

5.5.2 Sprint 2

In Sprint 2, we focused on designing the attendance tracking and leave management features in Figma. The completed designs include:

- Attendance dashboard UI
- Daily attendance record table (Employee, Department, Check-in/out, Hours, Status)
- Attendance summary cards (Present, Absent, Late, On Leave)
- Leave request page UI
- Leave balance cards (Annual, Sick, Business Leave)
- New leave request form (modal)
- Leave request submission success state
- Leave request list with status tracking (e.g., Pending)

For functional requirements, these features support:

- Viewing attendance records and employee status (Present, Late, Absent)
- Monitoring daily workforce summaries
- Submitting leave requests with details (type, date, reason, duration)
- Sending leave requests for manager approval
- Viewing leave history and request status

For non-functional requirements, these features provide:

- A clear and structured layout for attendance and leave data
- Consistency with the previous sprint (UI/UX continuity)
- Easy-to-use forms with minimal input complexity
- Clear user feedback (e.g., success messages, status indicators)
- Improved usability for tracking employee activity and leave requests

By the end of Sprint 2, the system was extended to include attendance monitoring and leave request workflows, enabling users to efficiently manage daily attendance and submit leave requests. The design ensures clarity, usability, and seamless interaction within the system.

Mon and the kids

Home
Calendar
Employee
Attendance
Leave

Hello! Lily
Let's ...

Attendance

Present Today 32 Absent 4 Late Arrivals 6 On Leave 2

Daily Record Today This week This month

Employee	Department	Date	Check In	Check Out	Hours	Status
Jame Doran Developer	Engineering	5 Mar, 2020	08:45 AM	-	7 hr 15 min	Present
Emily Wong Analyst	Finance	5 Mar, 2020	08:55 AM	05:10 AM	7 hr 15 min	Present
Tom Lee Dep Lead	Engineering	5 Mar, 2020	09:12 AM	05:10 AM	7 hr 3 min	Late
Michael Jackson Dep Lead	Engineering	5 Mar, 2020	-	-	-	Absent

Log Out

Lily Huys
HR Officer - Human Resources
lily.huys@mon.com

Check In
Check Out

Schedule

- Check In 08:45 - 09:00
- Lunch Break 12:00 - 12:30
- Check Out 12:30 - 12:45

Picture 5.3 Attendance Record Page (By HR)

Mon and the kids

Home
Calendar
Employee
Attendance
Leave

Hello! Stephen
Let's get to work

Leave Request

Annual Leave Remaining this year 12 Sick Leave Remaining this year 3 Business Leave Remaining this year 2

REQUEST

Log Out

Stephen Curry
Developer
stephen.c@mon.com

Check In
Check Out

Schedule

- Check In 08:45 - 09:00
- Lunch Break 12:00 - 12:30
- Check Out 12:30 - 12:45

Picture 5.4 Leave Request Page

Mon and the kids

Hello! Stephen
Let's ...

THIEN

2

New Leave Request
Submit your leave for manager approval

Leave type: e.g. Annual Leave

Start date: dd/mm/yyyy End date: dd/mm/yyyy

Reasons: Briefly describe the reason for your leave.

Duration: Day Request half day only

Approval: Gigi Hadid, Manager - IT

Cancel Submit Request

Log Out

Schedule

Check In: 08:00 AM

Lunch Break: 12:00 - 12:30

Check Out: 05:00 PM

Announcements

System maintenance: 17 Jan

19 February

Company holiday: 17 March

Picture 5.5 Leave Request Form

Mon and the kids

Hello! Stephen
Let's ...

THIEN

Annual Leave: 12, Sick Leave: 3, Business Leave: 2

REQUEST

✓ Leave Request Submitted
Your request has been sent to your manager for approval.

Log Out

Schedule

Check In: 08:00 AM

Lunch Break: 12:00 - 12:30

Check Out: 05:00 PM

Picture 5.6 Leave Request Submitted

The screenshot displays a web application interface for a user named Stephen. The browser address bar shows "www.url.com". The application has a sidebar on the left with a logo "Mon and the kids" and navigation links: Home, Calendar, Employee, Attendance, and Leave (highlighted). The main content area is titled "Hello! Stephen" and "Let's ...". Below this is a "Leave Request" section with three cards: "Annual Leave" (12 days remaining), "Sick Leave" (3 days remaining), and "Business Leave" (2 days remaining). A "REQUEST" button is visible. Below the cards, a pending leave request is shown for "Annual Leave" on "10 Mar 2025" (1 day - Full day), submitted by "Ogh Hadd" (Manager - IT) 1 day ago. The status is "Pending". On the right, there is a user profile for "Stephen Curry" with a "Check In" and "Check Out" button. Below the profile is a "Schedule" section showing a timeline with "Check In" at 08:00, "Lunch Break" from 12:00 to 13:00, and "Check Out" at 17:00.

Mon and the kids

Home

Calendar

Employee

Attendance

Leave

Log Out

Hello! Stephen
Let's ...

THEN

Leave Request

Annual Leave Remaining this year	12	Sick Leave Remaining this year	3	Business Leave Remaining this year	2
-------------------------------------	----	-----------------------------------	---	---------------------------------------	---

+ REQUEST

Annual Leave
10 Mar 2025
1 day - Full day
Ogh Hadd
Manager - IT
1 day ago
Pending

Check In
08:00

Lunch Break
12:00 - 13:00

Check Out
17:00

Picture 5.7 Pending Leave Request Form